

The reason for that was, primarily because of the very strict EMC immunity and emission requirements for automotive in-vehicle networking," he says. To overcome this, Broadcom has developed a technology called BroadR-Reach, which is essentially the ability to send and receive data simultaneously over a single pair of unshielded twisted pair cables, and to still be able to meet all of the automotive EMC quality and reliability standards.

"Wireless connectivity, including Wi-Fi and Bluetooth technology, is being rapidly adopted for automotive applications. Industry analysts have forecast that 100 per cent of vehicles will be connected by 2025 and there will be a 41 per cent increase in automotive wireless demand from 2012 to 2018. Wireless solutions will enable Miracast (a peer-to-peer wireless screen-casting standard) bandwidth for multiple in-car displays, smart remote and V2X connectivity and communication with biometric sensors and wearable technology," says Lau.

"In recent years, we have seen the addition of 2D/3D navigation systems, 3G/LTE wireless access, USB and Bluetooth connectivity. These systems are constantly connected to the car, to external devices and to the Internet," says Gupta of Freescale.

At this year's CES, automotive electronics was a key focus and many OEMs and car manufacturers launched varied technologies that will change the way we see automotive today. One such technology was Visteon's OASIS concept—an acronym for optimised, adaptable, secure, intelligent and seamless connectivity—that protects the vehicle communication network by providing a secure method to connect the vehicle to the user and cloud computing.

More updates.

► After 3G, it is going to be 4G in cars. GM announced it would be building a 4G/LTE connection into ten of its models.

► Audi also has similar plans for its A3 model, with AT&T as the data provider.

► Ford demonstrated its innovative vehicle-to-vehicle (V2V) communication

technology at CES 2014. It uses 802.11p, a flavour of Wi-Fi specific to the auto industry, to broadcast information such as the car's position, direction and speed to nearby cars, ten times per second. This information helps avoid collisions, amongst other benefits.

My car, my giant personal computer

What is connectivity without apps! Car makers are enabling users to make the most of the car's connectivity through a deep integration with mobile phones, as well as by offering dozens of smart apps through their own infotainment platforms. Going one step ahead, car makers are also linking wearable technology and auto features! Suppliers such as Delphi, Harman, QNX and Garmin are also demonstrating high-end infotainment systems for cars with app integration.

According to trends sighted at CES 2014, HTML5 is the programming language of choice for these platforms, as HTML5 makes it easier for third-party developers to create interesting new apps for cars. Likewise, a recent report by IHS Automotive shows that Linux is growing to be the preferred automotive infotainment platform. Models of Buick, GMC, Chevrolet and Opel use or will soon be using Linux. The report states that other car makers such as BMW, PSA and Jaguar/Land Rover are also committed to using a version of Linux that is compatible with the GENIVI platform.

More updates.

► BMW's iRemote App allows users to access vehicle information and control features from their mobile device and smart watch.

► Mercedes-Benz also recently showed how owners can use a smart watch to remotely unlock doors, check fuel levels, plan the car's trip or blow the horn.

► GM demonstrated its App Shop at CES 2014, providing the means for Chevrolet owners to download and install GM-approved apps right into their cars' dashboards.

► Audi's next-gen cars will feature

an in-built Android tablet, the 26cm (10.2-inch) Audi Smart Display. Powered by an NVIDIA Tegra 4 chip, the tab is extremely rugged with notably high resistance to temperature and physical damage. It uses the in-car Wi-Fi to provide Web access for entertainment, navigation and remote vehicle control.

► Mahindra e2o, the new electric car in India, has remote heating, ventilation, air-conditioning system (HVAC) activation via smartphone, a first in India. It also features telematics-based remote diagnostics, a first in India, and one of only four cars in the world and a best-in-class touch-screen-based infotainment unit with navigation, CAN bus integration, iPod integration, Bluetooth and EV status information.

My car, my silicon

Understandably, a smart car is packed with a myriad of electronics that implement everything from drive safety and engine management to car security and infotainment. This means big opportunity for the semiconductor industry.

"Semiconductor growth in the automotive market will be very significant. We are seeing a lot of growth, not just infotainment and ADAS, but also in powertrain, body control and safety electronics. All of those markets will see significant growth. There are a lot of different sectors that will be able to take a leading role in this market. Semiconductor providers who offer microcontrollers, communications technology and sensor technologies—all of those will take a leading role. At the same time, we believe that the Tier-1's, the resellers, developers, partners, all of them will take a leading role because the connected car ecosystem is going to require a lot more support in terms of software development, application development, hardware engineering, wiring suppliers, cabling suppliers, who will take a leadership role to help take that technology forward," says Lau.

The auto industry consumes microcontrollers, microprocessors, analogue ICs, discrete electronics, logic circuits, sensors and memory for electronic